Team Arjuna · QC-02 Pitch Script

Chakravyuh hackathon · Oct 27, 2026 · @Anthrosystemics

30-second elevator pitch

Say this when someone stops at the table, or as the opening line if time is cut.

> Nearly half the nitrogen in your body passed through one factory process, and that process uses 1–2% of global energy, because nitrogen's triple bond is brutally hard to break. Designing better catalysts needs energies accurate to 1.6 milli-hartree. The classical gold standard, CCSD(T), misses that even at equilibrium, then collapses below the true energy once the bond stretches. We built a verified pipeline: it shrinks the problem from 12 to 7 qubits, reaches chemical accuracy at all 12 bond lengths with ADAPT-VQE, shows honestly that today's noise wrecks that circuit, and shows that with IBM's sample-based quantum diagonalisation a noisy chip picks the important electron arrangements about 24 times better than chance on the stretched bond.

Full pitch script (about 6 minutes)

The same text is built into the deck: press N on any slide to see it. Final version for the 12-slide deck. Speaker A takes slides 1–5 (the problem and the idea), speaker B takes 6–12 (the build). Total about 6 minutes.

Slide	Who · time	Say this	Stage cue
1 · Title (H)	A · 0:20	Hi, we're Team Arjuna. Arjuna was the archer who saw only the bird's eye, and our target is one number: 1.6 milli-hartree. QC-02 asks us to compute the energy of a strongly correlated molecule to chemical accuracy, even on noisy hardware. Our molecule is nitrogen, N≡N.	Point at the arrow through the bond.

Slide	Who · time	Say this	Stage cue
2 · The problem (He)	A · 0:35	Air is 78% nitrogen, but plants can't use it until its triple bond is broken. Factories do that with Haber–Bosch at 300 to 500 degrees, using 1 to 2 percent of the world's energy; nearly half the nitrogen in your body went through it. An enzyme does the same job at room temperature. Designing catalysts like it needs bond energies within 1.6 milli-hartree, because at room temperature a 1.4 kcal/mol error already makes a predicted reaction rate ten times wrong.	Land on the pink 1.6.
3 · Why it's hard (Li)	A · 0:35	Electrons push on each other, so you can't solve them one at a time. An exact answer tracks every way they can sit in the orbitals: 400 for our model, 4.3 billion for nitrogen in a realistic basis, around 10^{30} for the enzyme's core. At rest one arrangement explains 93% of the molecule; stretched, the best explains 20%. That's static correlation, and it breaks the standard methods.	Pause on the two pies.

Slide	Who · time	Say this	Stage cue
4 · Gold standard fails (Be)	A · 0:35	CCSD(T) is the method chemists call the gold standard. Compared with the exact answer: 2.2 milli-hartree off at rest, already over budget; almost 11 off at 1.5 ångström; from 1.9 ångström it collapses 64 to 280 below the true energy, which no correct method can do, and at some lengths it doesn't converge at all. It breaks exactly where the chemistry gets interesting.	Trace the orange line up, then point at the pink triangles.
5 · Quantum in 60 s (B)	A · 0:35	A bit is 0 or 1; a qubit can be a blend of both until you measure it. Give every electron slot its own qubit: 1 means an electron is there. Twelve qubits hold a blend of all 4,096 patterns at once, the same crowd we just saw. Each measurement gives one pattern; we repeat thousands of times to estimate the energy. Over to [B].	Sweep across the three panels, then hand over.
6 · Our solution (C)	B · 0:30	Seven steps: a rough sketch of the electrons; zoom in on the six orbitals that make and break the bond; translate electron rules into qubit rules; delete qubits symmetry already fixes; build and tune the circuit with ADAPT-VQE; stress-test it under realistic noise; rescue the answer with SQD. Every step is checked against a known number before the next runs.	Follow the arrows with your hand.

Slide	Who · time	Say this	Stage cue
7 · Zoom, translate, shrink (N)	B · 0:30	Slide 7, nitrogen, and nitrogen's own orbitals. Only six take part in the bond, so we keep those and freeze the rest. Jordan–Wigner rewrites each electron move as qubit operations, 247 of them. Five symmetries, the electron counts and the molecule's mirror symmetries, fix five qubits, so we remove them exactly: 12 qubits become 7, CNOTs halve from 1,856 to 888, and the energy agrees to a billionth of a hartree.	Point at the crossed-out wires.
8 · ADAPT-VQE (O)	B · 0:40	Hot and cold: the chip makes a guess and measures its energy, a normal computer turns the knobs, and quantum mechanics guarantees lower energy is always a better guess. ADAPT grows the circuit one brick at a time, always adding the one that helps most. It lands on the exact curve at all 12 bond lengths, including where CCSD(T) failed. Stretched bonds need more bricks: 20 at rest, 35 stretched.	Point at the squares on the curve.

Slide	Who · time	Say this	Stage cue
9 · Noise and rescue (F)	B · 0:50	Now the honest part. Real two-qubit gates fail about once every 300 to 1,000 operations, and our accurate circuit has 1,000 to 1,800. We simulated that noise and tried the standard fixes: after readout correction and zero-noise extrapolation it was still more than 1.1 hartree off, and error cancellation would need 10^{13} repeats or more. Mitigation only reaches the target once gates are 30 to 100 times better than today. [pause] So we changed the chip's job. With SQD it only points at the arrangements that matter, and a normal computer solves exactly inside them. The fair test is a fixed budget: on the stretched bond, solving in a quarter of the space, the chip's picks are 9 milli-hartree off and random picks 227, so the chip carries about 24 times more information than chance. A space this small can eventually be covered by random sampling, so the payoff is at scale: IBM has run this method on nitrogen with 58 qubits.	Pause after "1.1 hartree off". This is the twist.

Slide	Who · time	Say this	Stage cue
10 · Tech stack (Ne)	B · 0:35	Left: our prototype, Python with NumPy and SciPy, written and checked by us; it produced every number in this deck. Right: the production tools in the same roles at bigger scale: PySCF for chemistry, qiskit-nature for translation, Qiskit and IBM's SQD add-on for algorithms, Aer and IBM Runtime for noise and real hardware, a Streamlit dashboard with automated tests. Each prototype module stays as the cross-check for the tool that replaces it.	Sweep left to right.
11 · Proof and next (Na)	B · 0:35	Every stage matches a published or exact number, and all 66 automated checks pass. Up front: the prototype uses a minimal basis, gate counts are upper bounds, and we haven't run on real hardware yet. During this hackathon: a realistic basis, real IBM noise models, IBM's SQD and one hardware run. After that: proper embedding and bigger molecules like iron–sulfur clusters, and long term, catalyst questions like the nitrogenase enzyme on early error-corrected machines.	Say the limits before a judge asks.

Slide	Who · time	Say this	Stage cue
12 · Thank you (Mg)	B · 0:10	To sum up: 12 qubits down to 7, chemical accuracy at all 12 bond lengths, and a noisy chip that, on the stretched bond, picks arrangements 24 times better than chance. Thank you; we'd love your questions.	Stop talking and smile.

Judge Q&A prep

The review panel was a quantum chemist, an IBM-style quantum engineer, an error-mitigation specialist and a VC. Each critiqued the deck, the answers were rewritten, and they critiqued again until no question was left without a short, true answer. Learn the first sentence of each answer; the rest is backup.

Likely question	Answer
STO-3G is a toy basis. Does any of this transfer?	Absolute energies don't, method gaps do. CCSD(T)'s failure at stretched N_2 is well known in realistic bases too. Porting to cc-pVDZ with PySCF is our first hackathon task.
Isn't a 12-qubit problem easy classically?	Yes, that's deliberate: exact answers let us grade every method. The point is the trend, which explodes to 4.3 billion arrangements in a realistic basis and about 10^{30} for FeMoco.
So is there a quantum advantage here?	Not at this size, and we don't claim one. We show where classical shortcuts fail, what a quantum method needs to fix it, and the hardware quality that requires.
Why ADAPT-VQE and not plain UCCSD?	ADAPT only adds operators that lower the energy, so circuits stay much shallower. We pick from 27 symmetry-allowed excitations instead of all 117.

Likely question	Answer
How exactly does tapering remove qubits?	Five Pauli Z-strings commute with H: electron-count parities and mirror symmetries. A Clifford rotation turns each into a single-qubit X with a known value, so that qubit is replaced by ±1 (Bravyi et al., 2017).
Your circuit has ~1,000 CNOTs. Isn't that unrealistic?	Yes, and that's our finding. The counts are upper bounds without circuit optimisation or routing; optimised excitation circuits cut them about 2–3×, still too deep for today's noise.
Why didn't ZNE fix it?	ZNE extrapolates a trend. When the signal is mostly gone, there is no trend left: at 3×10^{-3} error the circuit output is near random, so every fit lands far off. We amplify every gate, one- and two-qubit; ZNE reaches 1.6 mHa only below about 3×10^{-5}–10^{-4}.
Isn't PEC unbiased?	It is, but its cost grows exponentially with circuit size: about 10^{13}–10^{18} shots here. Unbiased but impossible to run.
Isn't your SQD result trivial, since the space only has 3,136 configurations?	Uncapped, yes: about 300 random bitstrings cover it, and we say so. That's why we test at a fixed budget against random and classical picks: on the stretched bond the chip is 24× better than random and 12× better than a simple classical ranking at 25% of the space. At equilibrium the classical ranking wins. IBM's paper shows the same idea at 58–77 qubits.
What's your noise model? Is it realistic?	Depolarising gate errors plus readout flips, at order-of-magnitude current values. It's a stand-in: next we load IBM backend calibrations into Qiskit Aer.
Did you run on real hardware?	Not yet. One shallow SQD run on an IBM Heron chip via the Open Plan is on our hackathon plan.

Likely question	Answer
How do we know your numbers are right?	Every stage matches a published or exact value: water's textbook Hartree–Fock energy to 3×10^{-13} Ha, Crawford's CCSD(T) to 10^{-10}, tapered vs untapered to 10^{-9}. One command reruns all 66 checks.
Who would pay for this? (VC)	Catalyst and materials R&D teams that already spend on simulation: ammonia, fuels, batteries. Near term the product is a verified benchmarking pipeline; long term it's quantum-accelerated catalyst screening.
What's "active-space embedding" here exactly?	An active space on a frozen mean-field core (CASCI): the six bond orbitals are solved exactly, the rest are frozen as a mean-field background. True embedding (DMET or projection-based) is on our roadmap.
Why N_2 and not FeMoco directly?	N_2 is the smallest system with the same failure mode, and it has exact answers to grade against. FeMoco needs about 108 qubits and fault tolerance.
How can CCSD(T) go below the exact energy?	Coupled cluster isn't variational, so nothing stops it undershooting. When the single-reference assumption fails at stretched bonds, it overshoots by 64 to 280 mHa below the exact answer, a well-known failure for N_2. A variational method like ADAPT-VQE can't go below the true energy.
Are you sure you're getting the right spin state?	Yes. We add a spin penalty, $H + 0.5\cdot S^2$, so ADAPT stays in the singlet; final $\langle S^2\rangle$ is below 3×10^{-4} at every length. At 3.0 Å the lowest state of any spin is actually a septet, so without the penalty you'd get the wrong state.
Isn't (T) zero in your 6-orbital active space?	Yes: with only three occupied and three virtual orbitals of the right symmetry the triples correction vanishes. That's why the headline benchmark uses the larger 10-electron, 8-orbital valence space, where (T) is non-zero.

Likely question	Answer
Can I try it myself?	Yes: Arjuna Lab is a single web page with the engine inside. Pick a bond length, build the circuit brick by brick, turn the noise up, and run SQD against random picks. It verifies itself against our Python numbers when it loads.

Jargon cheat sheet

If a judge uses a word you don't know, find it here first.

Term	Plain meaning
Hartree (Ha), milli-hartree (mHa)	The atomic unit of energy. 1 mHa is a thousandth of one.
Chemical accuracy	Error within 1 kcal/mol, about 1.6 mHa. Enough to predict which reaction wins.
Orbital	A region where an electron can sit. Each holds at most two electrons, one spin up and one down.
Configuration / determinant	One specific way of placing all electrons in the orbitals.
Hartree–Fock (HF)	The best single configuration; each electron feels only the average of the others.
Static (strong) correlation	Several configurations matter at once, so single-configuration methods break. Happens when bonds stretch.
CCSD(T)	Coupled cluster with singles, doubles and estimated triples. The classical "gold standard", built on one configuration.
Active space, CAS(6e,6o)	Solve exactly only 6 electrons in the 6 most important orbitals; freeze the rest.
Embedding	Treat a small important region with an expensive method inside a cheaper description of the rest.
Qubit	A quantum bit that can be a weighted blend of 0 and 1 until measured.

Term	Plain meaning
Superposition	That blend of possibilities.
Entanglement	Qubits linked so the whole state can't be described one qubit at a time.
Shot	One run of the circuit plus a measurement. Averages need thousands.
Jordan–Wigner	A rulebook that turns electron operations into qubit operations (Pauli strings).
Pauli strings (X, Y, Z)	Products of basic single-qubit operations; the Hamiltonian is a weighted sum of them.
Hamiltonian (H)	The energy operator. Its lowest value is the ground-state energy we want.
Z_2 tapering	Removing qubits whose values are fixed by symmetry. Exact, no approximation.
VQE	Variational quantum eigensolver: the chip measures energy, a classical optimiser tunes the circuit.
Variational principle	Any trial state's energy is at or above the true lowest energy, so lower is always better.
ADAPT-VQE	VQE that grows its circuit one operator at a time, picking the one with the biggest energy gradient.
CNOT	The main two-qubit gate. The noisiest operation, so we count them.
Noise / gate error	Chance that a gate does the wrong thing. Today roughly 10^{-3} for two-qubit gates.
Readout error mitigation (REM, TREX)	Correcting biased measurement results.
ZNE	Zero-noise extrapolation: run at 1×, 3×, 5× noise and extrapolate back to zero.
PEC	Probabilistic error cancellation: undo noise on average; exact but needs exponentially many shots.
Dynamical decoupling (DD)	Extra pulses on idle qubits to cancel slow drift.

Term	Plain meaning
SQD	Sample-based quantum diagonalisation: the chip samples important configurations, a classical computer solves exactly inside them.
Fault tolerance, logical qubit, code distance	Many physical qubits encode one error-corrected logical qubit; bigger distance means fewer errors.
T gate	The expensive gate in fault-tolerant machines; rotations are built from many of them.
DMRG	A strong classical method for larger active spaces, used as a reference.
FeMoco	The iron–molybdenum core of nitrogenase, the enzyme that breaks N≡N at room temperature.

Honesty rules

Judges forgive limits you state yourself; they punish limits they discover. Say "we built and verified" only for things in the first table.

Built and verified (say it confidently)	Not done yet (say "next")
Own integrals, Hartree–Fock, CCSD(T), exact solver, matched to published values	PySCF with a realistic basis (cc-pVDZ)
Jordan–Wigner mapping and tapering, 12 → 7 qubits, exact	qiskit-nature, Qiskit and Aer integration
ADAPT-VQE within 1.6 mHa at all 12 bond lengths (noiseless simulation)	Any run on real IBM hardware
Noise model with readout fix, ZNE and PEC cost, simulated	IBM backend calibration noise, dynamical decoupling
SQD at a fixed budget: noisy chip 24× better than random picks (stretched bond, simulated)	IBM's qiskit-addon-sqd at scale
66 automated checks pass (14 Python + 52 browser engine)	Streamlit dashboard, CI

Lines to use:

- "Everything on screen comes from runs of our own verified code. The production libraries are what we're integrating this week."
- "We don't claim quantum advantage at this size. We show where the classical method fails and what it takes to beat it."
- "The noise result is a finding, not a failure: it tells you the gate quality chemistry needs."

Sources: Haber process (Wikipedia) for energy share and the body-nitrogen figure · Robledo-Moreno et al., arXiv:2405.05068 for IBM's SQD on N_2 with 58 qubits · Reiher et al., 2017 for FeMoco's 54-electron, 54-orbital active space.